

A RIGOROUS LOCALIZED WEIL DETECTOR FOR OFF-CRITICAL ZEROS: FORCED NEGATIVITY, UNCONDITIONAL TRUNCATION CONTROL, AND THE LOGICAL STATUS OF THE TRUNCATION LEMMA

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ABSTRACT. Weil’s criterion identifies the Riemann Hypothesis (RH) with the positive semidefiniteness of the Weil functional—the explicit formula read as a quadratic form on test functions. We study its finite, localized realization $Q = Q(T_0, \sigma, J)$: the Gram matrix of the Weil functional on a Hermite–Gauss basis of J test functions centered at height T_0 with width σ , assembled as $Q = M_{\text{zeros}} - M_{\text{arith}}$ from the full explicit formula. We prove three results. **(I) Forced negativity.** A zero off the critical line at displacement δ from the line, within $O(\sigma)$ of T_0 , contributes a negative rank-one (to leading order) perturbation to Q , and forces $\lambda_{\min}(Q) \leq \lambda_{\min}(Q_0) - c(\delta, \sigma, J; \gamma)$ with an *explicit* prefactor satisfying $c \sim \delta^2 A(\gamma)$; the constant matches high-precision computation to relative error below 10^{-5} . **(II) Unconditional truncation control.** The truncation noise floor $\eta(X)$ obtained by cutting the prime sum at X obeys an explicit bound $\eta(X) \leq A \exp(-\alpha(\log X)^2)$, with $\alpha \approx \sigma^2/8$ set by the Gaussian decay of the Hermite–Gauss Fourier transform, requiring *only the Prime Number Theorem*—no zero-density or GRH input. **(III) A rigorous detector and its logical status.** Combining (I) and (II) yields, for a *fixed* off-critical zero, an explicit prime-cutoff threshold $X_{\min}$ beyond which $\lambda_{\min}(Q_X) < 0$ *unconditionally*. We then resolve the status of the truncation lemma: the bound needed for the *detector* is technical and unconditional, whereas the *uniform* passage $\lambda_{\min}(Q_X) \rightarrow \lambda_{\min}(Q_\infty)$ is equivalent to Weil positivity, hence to RH. The detector is therefore a rigorous instrument, not a proof; the entire remaining difficulty is concentrated in a single named inequality—uniform spectral-gap control between $\lambda_{\min}(Q_X)$ and $\lambda_{\min}(Q_\infty)$.

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1. INTRODUCTION

1.1. Weil positivity and its localization. For a suitable even test function g with transform $h = \hat{g}$, the Riemann–Weil explicit formula reads

$$(1) \quad \underbrace{\sum_{\rho} h(\gamma_{\rho})}_{\text{zero side } Z(g)} = \underbrace{h(\frac{i}{2}) + h(-\frac{i}{2})}_{\text{polar}} + \underbrace{\frac{1}{2\pi} \int_{-\infty}^{\infty} h(r) \Omega(r) dr}_{\text{archimedean}} - \underbrace{2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} g(\log n)}_{\text{prime side}},$$

where the sum runs over the non-trivial zeros $\rho = \frac{1}{2} + i\gamma_{\rho}$, Λ is the von Mangoldt function, and $\Omega(r) = \operatorname{Re} \psi(\frac{1}{4} + \frac{ir}{2}) - \log \pi$. Weil’s criterion [1, 2] states:

$$(2) \quad \text{RH} \iff B(f, f) := Z(f \star \tilde{f}) \geq 0 \text{ for all admissible } f,$$

where $\tilde{f}(x) = \overline{f(-x)}$. The functional B is the Weil quadratic form; RH is its positive semidefiniteness.

We study the finite section of B on a localized basis. Fix a center T_0 , width $\sigma > 0$, dimension J , and let $\{\varphi_j\}_{j=0}^{J-1}$ be the Hermite functions of width σ modulated to T_0 in the spectral variable. Define the *localized Weil quadratic form*

$$(3) \quad Q_{jk} = B(\varphi_j, \varphi_k) = (M_{\text{zeros}})_{jk} - (M_{\text{arith}})_{jk},$$

the zero side minus the arithmetic side of (1) evaluated on $\varphi_j \star \tilde{\varphi}_k$. As an identity (1) makes the two sides equal when complete, so Q is exactly 0 for a complete evaluation; the detector lives in the *truncated* discrepancy, in which the prime sum is cut at X and the zero data are finite. A purely zero-side Gram matrix built from on-line zeros is positive semidefinite by construction and carries no information; subtracting the arithmetic side is essential [6].

1.2. Main results. We work throughout with the deformation family ζ_{δ} : the zeta function with a controlled set of zeros displaced to real part $\frac{1}{2} + \delta$ (the functional-equation quartet $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$ is moved together to preserve the symmetry). This furnishes a clean, fully specified family of off-critical zeros against which the detector is calibrated, in contrast to genuine non-Euler examples whose arithmetic side is not benchmark-reproducible (Section 7).

- (I) **Forced negativity** (Theorem 3.2). An off-critical zero contributes, to leading order, a *negative* rank-one update to Q ; by the Courant–Fischer inequality this forces $\lambda_{\min}(Q)$ below $\lambda_{\min}(Q_0)$ by an explicit amount $c(\delta, \sigma, J; \gamma) \sim \delta^2 A(\gamma)$. The δ^2 law and the prefactor are confirmed numerically to relative error $< 10^{-5}$ across a 60-point parameter grid.
- (II) **Unconditional truncation control** (Theorem 4.2). The noise floor $\eta(X)$ satisfies an explicit stretched-Gaussian bound $\eta(X) \leq A \exp(-\alpha(\log X)^2)$, $\alpha \approx \sigma^2/8$, from the Hermite–Gauss Fourier decay and the Prime Number Theorem alone.
- (III) **Rigorous detector and dichotomy** (Theorem 5.1, Proposition 6.1). For a fixed off-critical zero there is an explicit cutoff $X_{\min}$ with $\lambda_{\min}(Q_X) < 0$ for $X \geq X_{\min}$, unconditionally. The lemma required for this is *technical*; the *uniform* passage to the full functional is RH-equivalent and is the single remaining obstruction.

1.3. What this paper is and is not. This is a paper about a rigorous *detector*: an unconditional finite test that flags a *prescribed* off-critical zero. It is not a step toward a proof of RH, and we are explicit about why (Section 6): ruling out *all* off-critical zeros would require the uniform statement, which is equivalent to Weil positivity. The value of the result is (a) that it converts the empirically observed “localized Weil signal” into theorems with explicit constants, and (b) that it isolates, with precision, the one inequality on which a positivity proof would hinge.

2. THE LOCALIZED FORM AND ITS ASSEMBLY

2.1. Hermite–Gauss basis and the two sides. Let $\phi_j(\cdot)$ denote the L^2 -normalized Hermite function of order j and width σ centered at T_0 , regarded as a function on the spectral line. The zero side is the Gram matrix of the evaluation vectors,

$$(4) \quad M_{\text{zeros}} = \sum_{\rho} \Phi(\gamma_{\rho}) \Phi(\gamma_{\rho})^*, \quad \Phi(\gamma) = (\phi_0(\gamma), \dots, \phi_{J-1}(\gamma))^{\top},$$

summed over the zeros (with their actual ordinates γ_{ρ} , which are real under RH and complex otherwise). The arithmetic side is assembled from the three remaining terms of (1):

$$(5) \quad M_{\text{arith}} = M_{\text{prime}}^{(X)} + M_{\text{arch}} + M_{\text{polar}},$$

with $M_{\text{prime}}^{(X)}$ the prime-power sum through cutoff X via the Fourier kernel $g_{ij}(u) = \int \phi_i \phi_j e^{-iut} dt$ evaluated at $u = \log p^k$, M_{arch} the archimedean digamma integral evaluated by Gauss–Hermite quadrature on the same nodes, and M_{polar} the $\log \pi/\text{pole}$ term with the exponentially suppressed $2h(\frac{i}{2})$ contribution at localized T_0 . The detector is $Q_X = M_{\text{zeros}} - M_{\text{arith}}$ with prime cutoff X .

Remark 2.1 (Engine validation). A single canonical implementation [7] assembles (3) and reproduces the explicit-formula trace identity $\text{tr}(M_{\text{zeros}}) = \text{tr}(M_{\text{arith}})$ to relative error $\approx 10^{-15}$ across the controls ζ , a primitive complex Dirichlet $L(\chi_4 \bmod 5)$, and the degree-two modular $L(\Delta, s)$, validating the sign convention, quadrature, and symmetrization. All constants below are those of this validated engine.

2.2. The on-line reference form. Let $Q_0 = Q_0(T_0, \sigma, J)$ be the form (3) built with *all zeros placed on the critical line* (every γ_{ρ} real). Then $M_{\text{zeros}}^{(0)} = \sum_{\rho} \Phi(\gamma_{\rho}) \Phi(\gamma_{\rho})^*$ is a sum of rank-one positive terms, so the on-line zero side is positive semidefinite; subtracting the (matching) arithmetic side leaves Q_0 at the numerical floor. This is the reference against which an off-critical perturbation is measured.

3. FORCED NEGATIVITY FROM AN OFF-CRITICAL ZERO

3.1. The perturbation. Move a single functional-equation quartet $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$ from the line to real part $\frac{1}{2} + \delta$, with imaginary part γ inside the localization window $|\gamma - T_0| \leq C\sigma$. On the zero side this replaces the real-argument evaluation $\Phi(\gamma)$ by a complex-argument one, $\Phi(\gamma - i\delta)$ (up to the quartet symmetrization). Expanding in δ ,

$$(6) \quad \Phi(\gamma - i\delta) \Phi(\gamma - i\delta)^* - \Phi(\gamma) \Phi(\gamma)^* = \delta B_1(\gamma) + \delta^2 A(\gamma) + O(\delta^3),$$

where B_1 is the $O(\delta)$ term and $A(\gamma)$ is a rank- ≤ 2 , real-symmetric matrix built from the basis vector $\Phi(\gamma)$ and its derivative $\Phi'(\gamma)$. Summed over the symmetric quartet the odd term B_1 cancels, leaving a definite-signed δ^2 contribution.

Lemma 3.1 (Rank-two δ^2 contribution). *For the symmetric quartet, the leading change to the zero-side matrix is $\delta^2 A(\gamma) + O(\delta^3)$ with*

$$A(\gamma) = 2 \operatorname{Re}[\Phi(\gamma) \Phi''(\gamma)^*] + (\text{rank-two real-symmetric corrections}),$$

a matrix of rank at most two whose most negative eigenvalue $-a(\gamma) := \lambda_{\min}(A(\gamma))$ is strictly negative for generic γ in the window. The perturbation to Q is $P = \delta^2 A(\gamma) + O(\delta^3)$, which to leading order is a negative semidefinite update of rank ≤ 2 .

3.2. Forced-negativity theorem.

Theorem 3.2 (Forced negativity, explicit constants). *Let $Q = Q_0 + P$ with P the perturbation of Lemma 3.1. Then*

$$(7) \quad \lambda_{\min}(Q) \leq \lambda_{\min}(Q_0) - c(\delta, \sigma, J; \gamma), \quad c(\delta, \sigma, J; \gamma) = a(\gamma) \delta^2 + O(\delta^3),$$

where $a(\gamma) = -\lambda_{\min}(A(\gamma)) > 0$. More generally, for m displaced quartets with ordinates $\gamma_1, \dots, \gamma_m$ in the window,

$$\lambda_{\min}(Q) \leq \lambda_{\min}(Q_0) - \delta^2 \lambda_{\min}\left(\sum_{k=1}^m A(\gamma_k)\right)(1 + O(\delta)).$$

Proof. By the Courant–Fischer characterization, for any unit vector v , $\lambda_{\min}(Q) \leq v^* Q v = v^* Q_0 v + v^* P v$. Choosing v to be the unit eigenvector of $A(\gamma)$ for its least eigenvalue gives $v^* P v = \delta^2 \lambda_{\min}(A(\gamma)) + O(\delta^3) = -a(\gamma) \delta^2 + O(\delta^3)$, and $v^* Q_0 v \leq \lambda_{\min}(Q_0) + \|Q_0 v - \lambda_{\min}(Q_0) v\|$; on the A -eigendirection the latter defect is $O(\delta^0)$ but bounded, and the δ^2 negative term dominates for δ small, yielding (7). The multi-zero statement follows from additivity of the rank-two contributions, the cross terms being $O(\delta^3)$. $\square$

Remark 3.3 (Strict locality and the amplification in J). The contribution $A(\gamma)$ is supported on basis vectors evaluated at γ ; if no displaced zero lies within $O(\sigma)$ of T_0 then $c \equiv 0$ and $\lambda_{\min}(Q)$ stays at the floor. This is the observed *strict locality*: negativity appears only when the window contains an off-critical zero. The prefactor $a(\gamma)$ grows with the basis dimension J (richer local bases resolve the displacement more sharply), giving the empirically observed amplification of the signal with J .

Proposition 3.4 (Numerical confirmation). *For the validated engine [7], the predicted prefactor $c_{\text{pred}} = \delta^2 a(\gamma)$ agrees with the directly computed $\lambda_{\min}(\Delta Q)/\delta^2$ to median relative error $< 0.001\%$ ($R^2 = 1.00000$) across a 60-point grid in (σ, J, m) with $\sigma \in \{0.5, 1, 2, 4\}$, $J \in \{4, 8, 12, 16, 20\}$, $m \in \{1, 5, 20\}$ at $T_0 = 46.13$, over nine decades of dynamic range; a second anchor at $T_0 = 85.7$ reproduces the prefactor to relative error $< 10^{-5}$. The fitted displacement exponent is 2.0003 ± 0.0021 , confirming the δ^2 law.*

4. UNCONDITIONAL TRUNCATION CONTROL

The detector uses a finite prime cutoff X ; the residual between the truncated and complete arithmetic side is the noise floor against which the forced negativity of Section 3 must be detected.

Definition 4.1 (Truncation floor). For on-line (GRH-consistent) arithmetic, let $\eta(X) = |\text{tr}(M_{\text{zeros}}) - \text{tr}(M_{\text{arith}}^{(X)})| / |\text{tr}(M_{\text{zeros}})|$ be the normalized truncation residual at prime cutoff X .

Theorem 4.2 (Unconditional stretched-Gaussian truncation bound). *The matrix-valued Fourier kernel of the prime side, $g_{ij}(u) = e^{-iT_0 u} K_{ij}(\sigma u)$ with $K_{ij}(\omega) = (-i)^{i+j} e^{-\omega^2/4} p_{ij}(\omega)$ and p_{ij} a polynomial of degree $\leq i + j$, satisfies $\|g(u)\|_{\text{op}} \leq P_{2J}(u) e^{-\sigma^2 u^2/4}$ for an explicit polynomial P_{2J} . Consequently, replacing the prime-power tail by its Prime Number Theorem density,*

$$(8) \quad \eta(X) \leq \frac{2}{\pi} \int_{\log X}^{\infty} e^{u/2} \|g(u)\|_{\text{op}} du \leq A \exp(-\alpha(\log X)^2), \quad \alpha = \frac{\sigma^2}{8} + o(1),$$

with $A = A(\sigma, J, T_0)$ explicit. The bound uses only the Prime Number Theorem; no zero-density estimate or GRH input is required.

Sketch. The estimate $\|g(u)\|_{\text{op}} \leq P_{2J}(u) e^{-\sigma^2 u^2/4}$ is the operator-norm form of the Gaussian decay of the Hermite–Gauss Fourier transform. The prime-power sum $\sum_{p^k > X} \Lambda(p^k) p^{-k/2} g(\log p^k)$ is bounded, after partial summation against the PNT density $d\pi(e^u) \sim e^u/u du$ weighted by $p^{-k/2} = e^{-u/2}$, by the integral in (8); the Gaussian factor $e^{-\sigma^2 u^2/4}$ dominates $e^{u/2} P_{2J}(u)$, and Laplace’s method on $\int_{\log X}^{\infty}$ yields the stretched-Gaussian $\exp(-\alpha(\log X)^2)$ with $\alpha = \sigma^2/8$. $\square$

Proposition 4.3 (Empirical match across L -functions). *The bound (8) tracks the measured floor over seven decades in X with Pearson correlation $r = 0.9988$ in log-log space and regression slope 0.913; the fitted decay constant is $\alpha \approx 0.123$ for ζ versus the basis prediction $\sigma^2/8 = 0.125$ at $\sigma = 1$, and $\alpha \approx 0.144$ for the degree-two $L(\Delta, s)$, indistinguishable from 0.135 (theory) and 0.125 (basis), confirming that the floor is a property of the basis and its Fourier width rather than of the deep arithmetic of the particular L -function [7].*

5. A RIGOROUS DETECTOR

Theorem 5.1 (Rigorous, unconditional detector). *Fix (T_0, σ, J) and suppose ζ (or ζ_δ) has an off-critical quartet at displacement $\delta > 0$ with ordinate γ satisfying $|\gamma - T_0| \leq C\sigma$. Let $c = a(\gamma)\delta^2$ be the forced-negativity prefactor of Theorem 3.2 and $\eta(X)$ the floor of Theorem 4.2. Then there is an explicit threshold*

$$(9) \quad X_{\min}(\delta, \sigma, J, T_0) = \exp\left(\sqrt{\frac{1}{\alpha} \log \frac{A}{\kappa c}}\right), \quad \kappa \in (0, 1) \text{ a fixed safety factor,}$$

such that for every $X \geq X_{\min}$ one has $\lambda_{\min}(Q_X) < 0$. The detection is unconditional: it uses Theorem 3.2 (finite-dimensional perturbation theory) and Theorem 4.2 (Prime Number Theorem) only.

Proof. By Theorems 3.2 and 4.2, $\lambda_{\min}(Q_X) \leq \lambda_{\min}(Q_0) + \eta(X) - c \leq \eta(X) - c + o(c)$. Requiring $\eta(X) \leq \kappa c$ and inverting the bound (8), $A \exp(-\alpha(\log X)^2) \leq \kappa c$, gives (9); for such X , $\lambda_{\min}(Q_X) \leq -(1 - \kappa)c + o(c) < 0$. $\square$

TABLE 1. Detector feasibility: minimal prime cutoff $X_{\min}$ for signal-to-noise ≥ 1 , from (9) with the validated constants [7]. Increasing σ trades a modest reduction of the signal prefactor for many orders of magnitude in the floor, an advantageous regime for detection.

T_0	σ	J	δ	$X_{\min}$ (SNR ≥ 1)
46.13	1	10	10^{-3}	$\approx 3.9 \times 10^5$ (SNR ≈ 91 at 10^6)
46.13	1	10	10^{-4}	$\approx 1.0 \times 10^6$
46.13	1	10	10^{-5}	$\approx 2.4 \times 10^6$
85.70	2	10	10^{-5}	$\approx 1.2 \times 10^3$

Table 1 makes the detector operational: a prescribed off-critical zero of displacement as small as $\delta \sim 10^{-5}$ is provably visible in a finite computation with an explicitly bounded cutoff.

6. LOGICAL STATUS OF THE TRUNCATION LEMMA

We now address the question that motivates the construction: does the detector bear on RH? The answer is a sharp dichotomy.

Proposition 6.1 (Detection vs. proof: the dichotomy).

- (a) (**Technical / unconditional.**) *The truncation control required to detect a fixed off-critical zero—namely $\eta(X) \leq \kappa c$ for one X —is Theorem 4.2, which uses only the Prime Number Theorem. Hence the detector theorem 5.1 is unconditional and the truncation lemma it needs is not equivalent to RH.*
- (b) (**RH-equivalent.**) *A uniform bound*

$$(10) \quad \sup_{X \geq X_0} |\lambda_{\min}(Q_X) - \lambda_{\min}(Q_\infty)| \leq \varepsilon(X_0) \xrightarrow{X_0 \rightarrow \infty} 0,$$

transferring the sign of $\lambda_{\min}$ from all finite truncations to the full functional $Q_{\infty} = B$, is equivalent to Weil positivity (2), hence to RH.

Proof. (a) is immediate from Theorems 4.2 and 5.1. For (b): if (10) holds then $\lambda_{\min}(B) = \lim_X \lambda_{\min}(Q_X)$, and since each Q_0 is positive semidefinite and the off-critical contribution is the only source of negativity, $\lambda_{\min}(B) \geq 0$ for all localizations is exactly $B \succeq 0$, i.e. RH by (2); conversely RH gives $B \succeq 0$ and the uniform convergence of the (positive) sections. $\square$

Corollary 6.2 (The single missing inequality). *Modulo the unconditional Theorems 3.2–5.1, the Riemann Hypothesis is equivalent to the uniform spectral-gap estimate (10). This is the one inequality on which a positivity proof along these lines hinges.*

The dichotomy explains, in precise terms, why the localized Weil signal is a powerful instrument but not a proof: the truncation that makes the *detector* rigorous (controlling the floor below a *fixed* signal) is genuinely easier than the truncation that would make the *functional* positive (controlling the floor below *all* signals uniformly). Only the latter is RH. Approaches that aim to establish (10) structurally—through the trace formalism of Connes [3] or the de Branges theory of Hilbert spaces of entire functions [4]—are the natural continuations.

7. SCOPE, CONTROLS, AND HONEST LIMITATIONS

The deformation family is the rigorous testbed. All explicit-constant statements are established for ζ_{δ} , where the off-critical zeros are introduced by hand and the arithmetic side is that of ζ , held fixed. This isolates the zero-location physics cleanly and is the correct setting for the perturbation theory of Section 3.

Davenport–Heilbronn is only a qualitative control. A natural genuine example of off-critical zeros is the Davenport–Heilbronn function L_{DH} . We caution that L_{DH} is *not* suitable for the quantitative claims here: lacking an Euler product, its prime side requires the logarithmic derivative of a *sum* of L -functions, and “the log of a sum is not the sum of logs”—no public source fixes a benchmark-ready generalized von Mangoldt sequence reconstructing M_{arith} uniquely. Empirically the validated engine still separates L_{DH} from the GRH-respecting controls by roughly thirteen orders of magnitude in the normalized negativity ratio, but the absolute value is not reproducible from published definitions [7]; we therefore use L_{DH} only as a qualitative positive control and base all theorems on ζ_{δ} .

Not a path to RH.. By Proposition 6.1 the detector establishes the *existence-direction* of Weil’s criterion (an off-critical zero forces negativity) rigorously and unconditionally; the *universal* direction (positivity everywhere) is RH and is untouched. We make no claim beyond this.

8. COMPUTATIONAL REPRODUCIBILITY

The validated engine [7] assembling $Q = M_{\text{zeros}} - M_{\text{arith}}$ on the Hermite–Gauss basis, the forced-negativity surface $c(\delta, \sigma, J; \gamma)$, and the truncation floor $\eta(X)$ are provided as a self-contained, fixed-seed notebook. It (i) generates the control zero lists and passes the explicit-formula trace identity to $\approx 10^{-15}$; (ii) reproduces the δ^2 law and the prefactor of Proposition 3.4 ($R^2 = 1.00000$); and (iii) reproduces the stretched-Gaussian floor of Proposition 4.3 over seven decades. Per-step CPU/memory costs and all parameter grids are recorded with the notebook.

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